Biology

Topic: Structural organisation in animals

Animal Tissues

1. Match the following columns and choose the correct option.

Column I		Column II	
1	Compound epithelium	a	Functions as a lining for body cavities, ducts, and tubes
2	Glandular epithelium	b	Limited role in secretion and absorption
3	Simple epithelium	c	Nuclei are located at the base
4	Columnar epithelium	d	Specialised for secretion

1) 1-b, 2-d, 3- a, 4-c
2) 1-c, 2-d, 3- a, 4-b
3) 1-b, 2-a, 3-d, 4-c
4) 1-c, 2- a, 3-d, 4-b

2. _____ epithelium is made of a single thin layer of flattened cells with irregular boundaries. Fill in the blank.

1) Squamous
2) Cuboidal
3) Columnar
4) Ciliated

3. Identify the incorrect statement from the following:

1) Simple epithelium is made of single layer of cells.
2) The compound epithelium consists of two or more cell layers and has protective function.
3) The cells are compactly packed with little intercellular matrix in epithelial tissue.
4) Ciliated epithelium is found in the walls of blood vessels and air sacs of lungs.

4. Consider the following statements:

(i) Goblet cells are the unicellular glands.
(ii) The products of exocrine glands are hormones and are secreted directly into the fluid bathing the gland.
(iii) Some of the squamous cells get specialized for secretion and are called glandular epithelium.
(iv) Compound epithelium is made up of more than one layer of cells and thus has a limited role in secretion and absorption.

Which of the above statements are correct?

1) (i) and (ii)

2) (ii) and (iii)
3) (iii) and (iv)
4) (i) and (iv)

5. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Free surface of columnar epithelium may have microvilli.
Reason (R): Microvilli help in secretion and absorption. In the light of the above statements, choose the correct answer from the options given below:
1) Both assertion and reason are true, and reason is the correct explanation of the assertion
2) Both assertion and reason are true, but reason is not the correct explanation of the assertion
3) Assertion is true, but reason is false
4) Both assertion and reason are false

6. Mucus, saliva, earwax, oil, milk and digestive enzymes are secreted by
1) Endocrine glands
2) Exocrine glands
3) Heterocrine gland
4) All of the above

7. The cell junctions that perform cementing to keep neighbouring cells together are
1) Gap junctions
2) Adhering junctions
3) Tight junctions
4) None of the above

8. Mark the incorrect statement w.r.t. epithelial tissue:
1) Has a free surface that faces either a body fluid or outside environment.
2) Cells are compactly packed with little intercellular matrix.
3) Forms covering or lining of some parts of the body.
4) Is highly vascular.

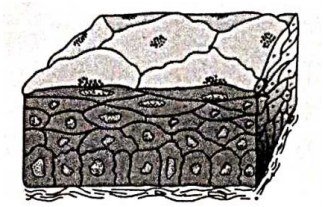
9. Which of the following cuboidal epithelial cells have microvilli to increase the absorptive surface area?
1) Fallopian tube
2) Proximal convoluted tubule
3) Capillary
4) Bowman's gland

10. _____ help to stop substances from leaking across a tissue.
1) Desmosomes
2) Tight junctions
3) Gap junctions
4) Adhering junctions

11. The cell junction specialized for rapid transfer of ions and molecules is
1) Adhering junction

2) Hemidesmosome
3) Gap junction
4) Tightjunction

12. Choose the option where the given epithelium is not found.



1) Inner lining of pancreatic ducts
2) Dry surface of skin
3) Moist surface of buccal cavity
4) Bowman's capsule

13. Gap junctions
1) Help to stop substances from leaking across a tissue.
2) Perform cementing to keep neighbouring cells together
3) Provide stretchability to the epithelium
4) Facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells

14. Read the following statements carefully:
1) Cuboidal epithelium is commonly found in duct of glands and tubular part of nephron.
2) Ciliated epithelium is found in inner lining of stomach and intestine.
3) Simple columnar epithelium is composed of a single layer of tall and slender cells.
4) Squamous epithelium is found in inner surface of bronchioles and fallopian tube.

Choose the option having incorrect statements.
1) (a) and (c) only
2) (b) and (d) only
3) (b), (c) and (d)
4) (a), (b) and (c)

15. Identify correct statements.
(a) In nearly all animal tissues specialised junction provide both structural and functional links between its individual cells.
(b) Gap junctions help to stop substances from leaking across a tissue.
(c) Tight junctions perform cementing to keep neighbouring cells together.
(d) Epithelial tissues are most abundant and widely distributed in the body of complex animals.
1) (a) and (d) only
2) (b) and (c) only
3) (a) only
4) (a), (c) and (d) only

16. Identify the incorrect statement w.r.t. bone.

1) Bone has hard and non-pliable ground substance rich in calcium salts and collagen fibres.
2) Bone is the main tissue that provides structural frame to the body.
3) The bone cells are present in spaces called lacunae.
4) All bones are the site of production of blood cells.

17. Mark the wrong statement about cartilage.

1) The intercellular material of cartilage is solid, pliable and resists compression.
2) Chondrocytes are enclosed in small cavities within the matrix secreted by them.
3) Cartilage serves weight-bearing function.
4) Most of the cartilage in vertebrate embryos are replaced by bones in adults.

18. Tendons and ligaments are examples of

1) Specialized connective tissue
2) Dense regular connective tissue
3) Dense irregular connective tissue
4) Loose connective tissue

19. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Muscle fibres contract (lengthen) in response to stimulation, then relax (shorten) and return to their uncontracted state in a coordinated fashion.

Reason (R): The wall of internal organs such as the blood vessels usually do not contract under our will.

In the light of the above statements, choose the correct answer from the options given below:

1) Both assertion and reason are true, and reason is the correct explanation of the assertion
2) Both assertion and reason are true, but reason is not the correct explanation of the assertion
3) Assertion is true, but reason is false
4) Assertion is false, but reason is true

20. Neuroglia are

1) Sensory cells
2) Secretory cells
3) Sensory and supporting cells
4) Non-sensory and supporting cells

Organ and Organ System

21. Each organ in our body is made up of

1) Organ system
2) Organism

3) Tissues
4) None of the above.

22. Our heart consists of
1) Epithelial tissue
2) Muscular tissue
3) Neural tissue
4) All of the above

23. Morphology refers to study of
1) Externally visible features
2) Internal organs
3) Evolutionary history
4) Heredity and variation

24. Anatomy refers to study of
1) External form
2) Internal organs
3) Animals
4) Both (1) and (3)

25. Organ system is absent in
1) Pigeon
2) Frog
3) Sponges
4) Cockroach

26. The complexity in organ and organ systems displays certain discernible trend. This discernible trend is called
1) Morphological trend
2) Evolutionary trend
3) Anatomical trend
4) Paleontological trend

27. Select the incorrect match.
1) Anatomy - Study of morphology of internal organs in the animals
2) Morphology - Study of form or externally visible features
3) Organ - Made of one or more type of organ systems
4) Tissues - Organise to form organs

28. Read the following statements and choose the incorrect option.
1) Organs associate to form organ systems in the multicellular organisms.
2) In case of animals, the term morphology refers to the external appearance of the organs or parts of the body.
3) In the case of plants or microbes, the. term morphology precisely means only study of form or externally visible features.
4) Organs are organised into tissues and tissues are orgahised into organ system, such an organisation is essential for more efficient and better coordinated activities of few cells constituting an organism.

29. Select the correct hierarchy in ascending order.

1) Tissue→Cell→Organ→Organ system→Organism
2) Cell→Tissue→Organ→Organ system→Organism
3) Organ→Cell→Tissue→Organ system→Organism
4) CellOrgan→Tissue→Organism→Organ system

30. An organ is usually made up of

1) Same type of cells
2) Same type of tissues
3) Different types of tissues
4) None of the above

Cockroach

31. In cockroach, wings are attached with

1) Prothorax and mesothorax
2) Mesothorax and metathorax
3) Prothorax and metathorax
4) Abdomen

32. In the gut of cockroach, hepatic caecae are present

1) At the junction of foregut and midgut
2) At the junction of midgut and hindgut
3) Attached to rectum
4) Between crop and gizzard

33. Cockroach is

1) Omnivorous
2) Found in damp places throughout the world
3) Vector of several diseases
4) All of the above

34. Identify the incorrect statement w.r.t. cockroach.

1) Head is connected to thorax by a short extension of prothorax known as the neck.
2) Forewings or tegmina are opaque, dark and leathery.
3) Forewings cover the hindwings while flying.
4) Hindwings are transparent, membranous and are used in flight.

35. Match the options given in column I with column II w.r.t. mouth parts of cockroach.

Column I		Column II	
A	Labrum	I	Tongue
B	Labium	II	Upper lip
C	Hypopharynx	III	Lower li

1) A-I, B-II, C-III
2) A-II, B-1, C-III
3) A-III, B–II, C–I

4) A-II, B-III, C-I

36. Which of the following is an incorrect statement w.r.t. circulatory system in cockroach?

1) Blood vascular system of cockroach is of open type.
2) Blood vessels are poorly developed and open into space (haemocoel).
3) Visceral organs located in the haemocoel are bathed in blood..
4) The haemolymph is composed of coloured plasma and erythrocytes.

37. The heart of cockroach is

1) Elongated muscular tube
2) 13 chambered
3) Lies along mid-ventral line of thorax and abdomen
4) Both (1) and (2)

38. Select the incorrect match.

1) Malpighian tubules - Yellow coloured thin filamentous
2) Tegmina - Cover the hind wings when at rest
3) Mouth parts - A labrum, a mandible, a maxilla and a labium
4) Hypopharynx - A median flexíble lobe

39. At the junction of midgut and hindgut of cockroach is present a ring of 100-150 yellow-coloured thin filamentous

1) Gastric caecae
2) Hepatic caecae
3) Malpighian tubules
4) Rectal papillae

40. Mark the incorrect statement regarding digestive system of cockroach.

1) Oesophagus opens into sac-like crop used for storing food.
2) Crop is followed by gizzard or proventriculus.
3) Gizzard has on outer layer of longitudinal muscles and thick inner cuticle forming eight highly chitinous plate like teeth.
4) Gizzard helps in grinding the food particles.

41. Identify the incorrect statement w.r.f. the morphology of cockroach.

1) Their size ranges from 0.6 to 7.6 cm.
2) Bright yellow, red and green coloured cockroaches are reported.
3) They have 2 pairs of antennae.
4) Flat extension of upper body wall conceals head.

42. The vision in cockroach is mosaic with

1) More resolution, but less sensitivity
2) More sensitivity, less resolution
3) Ommatidia that receives several images of an object
4) Both (2) and (3)

43. Each compound eye of cockroach consists of about _____ ommatidia

1) 2000 triangular
2) 1000 hexagonal

3) 2000 hexagonal
4) 2000 circular

44. In female cockroaches
1) The seventh sternum is boat-shaped
2) The seventh sternum together with eighth and ninth sterna forms a brood or genital pouch
3) The anterior part of the genital pouch contains female gonopore, spermathecal pores and collateral glands
4) All of the above

45. Read the following statements about cockroach.
(i) Head is triangular in shape and lies antériorly at right angles to the longitudinal axis.
(ii) Head is formed by fusion of nine segments and has restricted mobility.
(iii) The entire midgut of cockroach is lined by cuticle.
(iv) Antennae of cockroach has sensory receptors for monitoring the environment.
(v) Mushroom glands are present in 4th-6th abdominal segment in male cockroach.

Which of the above statements are correct?
1) (i) and (iii)
2) (ii) and (v)
3) (iv) and (v)
4) (i) and (iv)

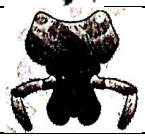
46. The female reproductive system of cockroach consists of two large ovaries lying
1) Dorsally in 4th-6th abdominal segments
2) Laterally in 2nd 6th abdominal segments
3) Laterally in 4th-8th abdominal segments
4) Ventrally in 6th_8th abdominal segments

47. Which of the following mouth parts of cockroach are unpaired?
1) Mandibles and labrum
2) Mandibles and maxillae
3) Maxillae and labium
4) Labrum and hypopharynx

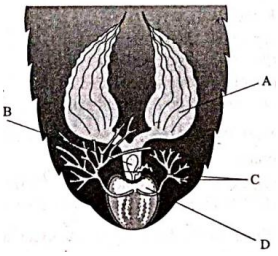
48. Given below are two statements:
Statement I: Each ovary in female cockroach is formed of a group of eight ovarioles, containing a chain of developing ova.
Statement II: Wings extend beyond the tip of the abdomen in female cockroach.
In the light of the above statements, choose the most appropriate answer from the options given below:
1) Both statement I and II are incorrect.
2) Statement I is correct, but statement II is incorrect.
3) Statement I is incorrect, and statement II is correct.

4) Both statement I and II are correct.

49. Mark the incorrect figure with its description:

	Column I	Column II
1)		Opening of common salivary duct on its basal part
2)		Upper lip with sensory setae
3)		Has five segmented maxillary palps
4)		Lower lip with sensory setae

50. Following is the diagram of reproductive system of. female cockroach with parts labelled A, B, C and D. Which of the following is an incorrect identification with its features/functions?



1) A-Pair of ovaries lying laterally in 2nd 6th abdominal segments
2) B-Pair of spermatheca in 6th segment
3) C-Phallic gland which secrets ootheca
4) D-Genital chamber which is a part of genital pouch in female

Frogs

51. Frog respires through _____ during hibernation and aestivation. Fill in the blank with the correct option.

1) Skin
2) Lungs
3) Buccopharynx
4) Gills

52. Male frogs can be distinguished from female due to the presence of

1) Copulatory pads
2) Darker colour
3) Both (1) and (2)
4) Nictitating membrane

53. The alimentary canal of frog is

1) Long since it is a herbivore
2) Short because it is a carnivore
3) Incomplete since anus is absent
4) Short because it is a herbivore

54. The heart of frog has
1) Two atria and two ventricles
2) Two atria and one ventricle
3) One atrium and two ventricles
4) One atrium and one ventricle

55. The glands absent in frog are
1) Gastric glands
2) Salivary glands
3) Cutaneous glands
4) Mucus glands

56. Female frog lays
1) 2500-3000 unfertilized eggs
2) 2500-3000 fertilized eggs
3) 3000-5000 unfertilized eggs
4) 3000-5000 fertilized eggs

57. Mark the odd one wr.t. paired structures found in frog's brain
1) Cerebral hemisphere
2) Optic lobes
3) Olfactory lobes
4) Diencephalon

58. Which of the following is a correct statement for Rana tigrina?
1) It drinks water on land.
2) It drinks water only when it is in ponds.
3) Frog never drinks water; it only absorbs it through skin.
4) Frog drinks more water in summers than in winters

59. Sinus venosus is an accessory chamber of heart attached with _____ in frogs. Fill in the blank with the correct option.
1) Right ventricle
2) Right atrium
3) Left ventricle
4) Left atrium

60. Which of the following is an incorrect statement?
1) Frog plays an important role in maintaining ecological balance.
2) Fertilization is external, and development is indirect in frog.
3) Frogs has 12 pairs of cranial nerves.
4) Frogs has both renal portal system and hepatic portal system.

61. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason(R).

Assertion (A): The alimentary canal of frog is short.

Reason (R): Frogs are herbivorous animals.

In the light of the above statements, choose the correct answer from the options given below:

1) Both (A) and (R) are correct but (R) is the not correct explanation of (A).
2) (A) is correct but (R) is not correct.
3) (A) is not correct but (R) is correct.
4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

62. In male frog, testes are adhered to the upper part of kidneys by a double fold of peritoneum called _____. Fill in the blank.

1) Mesovarium
2) Mesorchium
3) Mesosalpinx
4) Mesentery

63. In female frog, the ovaries

1) Are paired
2) Are situated near kidneys
3) Are functionally connected with kidneys
4) Both (1) and (2)

64. Match the following columns related to frog and choose the correct option.

Column I		Column II	
1	Forebrain	a	Cerebellum and medulla oblongata
2	Midbrain	b	Diencephalon and mesencephalon
3	Hindbrain	c	Olfactory lobes, cerebral hemispheres and diencephalon
		d	and diencephalon Optic lobes

1) 1-b, 2-d, 3-a
2) 1-c, 2-d, 3-a
3) 1-d, 2-b, 3-c
4) 1-b, 2-d, 3-c

65. RBCs of frog are

1) Biconvex and nucleated
2) Biconcave and nucleated
3) Biconvex and enucleated
4) Biconcave and enucleated

66. Identify the wrong statement w.r.t. frog.

1) Frogs exhibit sexual dimorphism.
2) In water, buccal cavity acts as aquatic respiratory organ.
3) Hindlimbs are larger and muscular than forelimbs.
4) Membranous tympanum receives sound signals.

67. Identify the correct set of statements:

(a) Frogs exhibit sexual dimorphism.

(b) Male frogs have copulatory pad on the first digit of the forelimbs.

(c) In male frogs the ureters and ejaculatory ducts open seperately in the cloaca.
(d) Frogs are ureotelic animals.
(e) External ear is absent in frogs and only tympanum can be seen externally.
Choose the correct answer from options given below:
1) a and d only
2) b, c, d and e only
3) a, b, d and e only
4) b and c only

68. Given below are two statements:
Statement I: Frog shows aestivation and hibernation.
Statement II: The female frogs can be distinguished by the presence of sound producing vocal sac which are absent in male frogs.
In the light of the above statements, choose the most appropriate answer from the options given below:
1) Both statement I and II are incorrect.
2) Statement I is correct, but statement II is incorrect,
3) Statement I is incorrect, and statement Il is correct.
4) Both statement I and II are correct.

69. Identify the correct set of statements:
(a) Above the mouth, a pair of nostrils is present.
(b) In frog, fertilization is internal
(c) Eyes in a frog are a pair of oval structures situated in the orbit in skull.
(d) In male frog number of vasa efferentia are 10-12.
(e) Endocrine glands are not found in frogs.
(f) Choose the correct answer from options given below.
1) a and d only
2) b, c, d and e only
3) a, b, d and e only
4) b and c only

70. Select the correct sequence for the passage of sperms in male frogs.
1) Testes→ Vasa efferentia → Kidney → Seminal vesicles → Urinogenital duct → Cloaca
2) Testes → Vaså efferentia → Kidney→ Bidder's canal → Urinogenital duct → Cloaca
3) Testes → Kidney → Vasa efferentia → Bidder's canal → Urinogenital duct → Cloaca
4) Testes→ Vasa efferentia → Bidder's canal Kidney → Urinogenital duct → Cloaca

Answer Keys

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
1	1	4	4	1	2	2	4	2	2	3	4	4	2	3
16	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
4	3	2	4	4	3	4	1	2	3	2	3	4	2	3
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**	**41**	**42**	**43**	**44**	**45**
2	1	4	3	4	4	4	3	3	3	3	4	3	4	4
46	**47**	**48**	**49**	**50**	**51**	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
2	4	2	2	3	1	1	2	2	2	1	4	3	2	3
61	**62**	**63**	**64**	**65**	**66**	**67**	**68**	**69**	**70**					
2	2	4	2	1	2	3	2	1	2					